

FoundationPose vs. EKF-Based Stereo Pose Tracking for Manipulator Visual Servoing

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Abstract—Robust visual servoing is critical for autonomous manipulation in unstructured environments such as warehouses and homes. Visual servoing requires accurate estimation of the relative pose between the robot end-effector and the target grasp point. We compare three approaches to pose estimation in an iterative visual servoing loop on real hardware: (i) a raw IBVS baseline that drives a calibrated image Jacobian directly from per-frame centroids, (ii) an EKF-filtered PBVS pipeline that fuses a learning-based perception front-end (DINOv2 + SAM2 + Depth Anything V2) into a temporally smoothed 3D position, and (iii) an EKF-filtered PBVS pipeline backed by FoundationPose for direct 6-DoF pose measurements. All three share the same xArm 7 manipulator, ZED Mini eye-in-hand camera, and per-frame CSV metrics logger, so differences are attributable to the perception/estimator stack. We describe the system architecture, the EKF formulation (including a Joseph-form covariance update, egomotion compensation, and a 2D-only partial-update fallback), and the experimental protocol used to quantify convergence speed, trajectory straightness, final error, and robustness under occlusion and lighting changes. Preliminary results are reported in `PLACEHOLDER` tables and will be updated once the full hardware sessions are complete.

I. INTRODUCTION

Visual servoing uses visual feedback to guide a robot toward a desired configuration. In applications like warehouse pick-and-place and household manipulation, the robot must localize a target object and iteratively correct its end-effector pose until it reaches a grasp point. The quality of this process depends directly on how well the perception system estimates the object’s pose at each control iteration. Errors in pose estimation do not simply add noise to the trajectory; they compound over time through the control loop, and can cause the robot to oscillate, overshoot, or diverge entirely from the desired grasp point.

There are two classical formulations for visual servoing [3]. Image-based visual servoing (IBVS) works entirely in pixel coordinates, mapping the error between current and desired image features to robot velocities through an image Jacobian. Because it avoids explicit 3D reconstruction, IBVS is simple and robust to calibration errors. However, IBVS can produce unintuitive Cartesian trajectories, struggles with large displacements, and has no way to reason about depth or distance to the target. Position-based visual servoing (PBVS) reconstructs the 3D pose of the target in the camera frame and computes a Cartesian velocity command that drives

the object to a desired position. PBVS gives straight-line trajectories in Cartesian space, but its performance is tied directly to the accuracy of the underlying 3D pose estimate.

Recent foundation models for perception open up new possibilities for building PBVS systems that work on novel objects without task-specific training. DINOv2 [5] provides dense, semantically rich patch-level features from a self-supervised Vision Transformer that transfer well across visual domains. SAM2 [6] produces high-quality segmentation masks from minimal prompts (a bounding box or a few points) and supports temporal mask propagation across video frames via logit passing. Depth Anything V2 [7] estimates monocular depth from a single RGB image, providing relative depth that can be calibrated to metric scale. FoundationPose [1] takes the next step and returns a full 6-DoF pose given RGB-D plus a mask plus either a CAD mesh or a handful of reference views. Together these models span the spectrum from light-weight monocular detection to dense stereo-based pose estimation, giving us three concrete perception back-ends to compare inside the same servoing loop.

However, foundation model outputs are inherently per-frame: each image is processed independently with no temporal context, so frame-to-frame jitter in the centroid or depth feeds straight into the control command and can easily produce oscillation near the goal. The Extended Kalman Filter (EKF) [2] addresses this by maintaining a probabilistic state (position and velocity) with a covariance matrix, which dampens measurement noise, propagates predictions forward during brief tracking failures, and naturally absorbs the robot’s own motion (egomotion compensation) since the camera is on the moving end-effector.

In this project, we build a complete visual servoing system that combines foundation model perception with EKF-based state estimation, and compare three full pipelines end-to-end: a raw IBVS baseline, an EKF-PBVS pipeline fed by DINOv2 + SAM2 + Depth Anything V2, and an EKF-PBVS pipeline fed by FoundationPose. All three run on real hardware (xArm manipulator, ZED Mini camera) with per-frame metrics logging for quantitative evaluation. Our goal is to characterize the practical trade-offs: does the EKF’s temporal filtering and uncertainty tracking provide meaningful improvements in convergence speed, trajectory smoothness, and final accuracy, and does that improvement depend on the quality of the underlying perception model?

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II. RELATED WORK

Visual servoing. Chaumette and Hutchinson [3] provide a comprehensive treatment of visual servo control, covering both IBVS and PBVS formulations, their stability properties, and the interaction matrix (image Jacobian) that relates image feature velocities to camera velocities. Their framework underpins most modern visual servoing systems. A key insight from their analysis is that IBVS is locally stable around the desired configuration but can fail for large displacements, while PBVS provides global convergence guarantees but is sensitive to pose estimation errors.

6-DoF pose estimation. FoundationPose [1] achieves state-of-the-art 6-DoF pose estimation and tracking from RGB-D input paired with either a CAD model or a small set of reference images. It has been benchmarked extensively on the BOP and YCB-Video datasets. However, it operates per-frame without temporal filtering, and it's reliance on a known 3D model or RGB-D input limits applicability in settings with novel objects or monocular cameras.

EKF for visual tracking. The use of Kalman filtering for visual tracking has a long history [2]. The standard approach is to maintain a state vector (typically position and velocity in 3D) and update it with measurements derived from image features. The EKF linearizes the nonlinear measurement model (pinhole projection) around the current estimate at each step. Tao et al. [4] recently proposed a perception-control coupled approach for visual servoing of textureless objects using a keypoint-based EKF. Their work is close in spirit to ours, but they use learned keypoint detectors as the measurement source, whereas we use foundation model features (DINOv2 + SAM2) and, separately, full FoundationPose translations.

Foundation models for robotics perception. DINOv2 [5] has been shown to produce patch-level features that transfer across visual domains for matching, retrieval, and segmentation tasks. SAM2 [6] extends the Segment Anything model with video-level mask propagation, enabling real-time tracking by passing low-resolution logits between consecutive frames. Depth Anything V2 [7] provides monocular depth estimation that, while producing relative rather than metric depth, can be converted to approximate metric scale through a simple affine calibration. These models have individually been applied to robotic tasks, but to our knowledge, no prior work has combined all three into an integrated perception front-end feeding an EKF for closed-loop visual servoing on real hardware and directly compared that stack against an EKF fed by a dedicated 6-DoF pose estimator in the same control loop.

III. THEORETICAL BACKGROUND

A. Pinhole Camera Model

A point $\mathbf{p} = [X, Y, Z]^T$ in the camera frame projects to pixel coordinates (u, v) via:

$$u = f_x \frac{X}{Z} + c_x, \quad v = f_y \frac{Y}{Z} + c_y \quad (1)$$

where f_x, f_y are the focal lengths in pixels and (c_x, c_y) is the principal point. The inverse operation (backprojection) given a known depth Z recovers the 3D point: $X = (u - c_x)Z/f_x$, $Y = (v - c_y)Z/f_y$.

B. Extended Kalman Filter

We use an EKF to track the target object's 3D position and velocity in the camera frame. The state vector is:

$$\mathbf{x} = [X, Y, Z, \dot{X}, \dot{Y}, \dot{Z}]^T \in \mathbb{R}^6 \quad (2)$$

where (X, Y, Z) is the object position in metres and $(\dot{X}, \dot{Y}, \dot{Z})$ is velocity in m/s, both in the camera coordinate frame.

Process model. We assume constant-velocity dynamics with egomotion compensation. Given time step Δt and the robot end-effector velocity $\mathbf{v}_{\text{robot}}$ expressed in the camera frame:

$$\mathbf{x}_{k|k-1} = \mathbf{F} \mathbf{x}_{k-1} - \begin{bmatrix} \mathbf{v}_{\text{robot}} \Delta t \\ \mathbf{0}_3 \end{bmatrix} + \mathbf{w}_k \quad (3)$$

where $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q} \Delta t)$ is process noise scaled by the time step, and the state transition matrix is:

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix} \in \mathbb{R}^{6 \times 6} \quad (4)$$

The egomotion subtraction accounts for the fact that when the camera (mounted on the end-effector) moves in direction $+X$, the object appears to shift in direction $-X$ in the camera frame. Without this compensation, the EKF would interpret robot-induced image motion as object motion and produce incorrect velocity estimates. The covariance prediction follows the standard form: $\mathbf{P}_{k|k-1} = \mathbf{F} \mathbf{P}_{k-1} \mathbf{F}^T + \mathbf{Q} \Delta t$.

Measurement model (pixel + depth). The measurement vector combines the 2D pixel centroid of the detected object with a metric depth estimate:

$$\mathbf{z} = \begin{bmatrix} u \\ v \\ Z \end{bmatrix}, \quad h(\mathbf{x}) = \begin{bmatrix} f_x X/Z + c_x \\ f_y Y/Z + c_y \\ Z \end{bmatrix} \quad (5)$$

This is the standard pinhole projection (Eq. 1) augmented with a direct depth observation. The measurement Jacobian $\mathbf{H} = \partial h / \partial \mathbf{x}$ evaluated at the current state estimate is:

$$\mathbf{H} = \begin{bmatrix} f_x/Z & 0 & -f_x X/Z^2 & 0 & 0 & 0 \\ 0 & f_y/Z & -f_y Y/Z^2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (6)$$

The velocity states $(\dot{X}, \dot{Y}, \dot{Z})$ do not appear directly in the measurement model, so the rightmost three columns of $\mathbf{H}$ are zero. The velocity is estimated indirectly: the process model couples position and velocity, and the position corrections from each measurement update propagate through the covariance to constrain the velocity as well.

Update step. Given a measurement $\mathbf{z}_k$ with noise covariance $\mathbf{R} = \text{diag}(\sigma_u^2, \sigma_v^2, \sigma_Z^2)$, we compute:

$$\mathbf{y}_k = \mathbf{z}_k - h(\mathbf{x}_{k|k-1}) \quad (\text{innovation}) \quad (7)$$

$$\mathbf{S}_k = \mathbf{H} \mathbf{P}_{k|k-1} \mathbf{H}^T + \mathbf{R} \quad (\text{innovation covariance}) \quad (8)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}^T \mathbf{S}_k^{-1} \quad (\text{Kalman gain}) \quad (9)$$

$$\mathbf{x}_k = \mathbf{x}_{k|k-1} + \mathbf{K}_k \mathbf{y}_k \quad (\text{state update}) \quad (10)$$

For the covariance update, we use the Joseph form rather than the standard $\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_{k|k-1}$, because the Joseph form is algebraically equivalent but numerically more stable when the Kalman gain is large:

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_{k|k-1} (\mathbf{I} - \mathbf{K}_k \mathbf{H})^T + \mathbf{K}_k \mathbf{R} \mathbf{K}_k^T \quad (11)$$

2D-only update. Running the depth model on every frame is expensive. On frames where we skip depth estimation, we perform a partial update using only the pixel centroid (u, v) . This uses the top two rows of $\mathbf{H}$ (a 2×6 matrix) and the corresponding 2×2 block of $\mathbf{R}$. The 2D-only update still constrains the lateral position (X, Y) while allowing the depth Z to evolve under the process model.

Direct 3D update. When the perception back-end is FoundationPose, the measurement is already a 3D position (X, Y, Z) in the camera frame, so the measurement model collapses to $h(\mathbf{x}) = [X, Y, Z]^T$ with $\mathbf{H} = [\mathbf{I}_3 \ \mathbf{0}_3]$ and $\mathbf{R}_{3d} = \sigma_{\text{pos}}^2 \mathbf{I}_3$. No projection Jacobian is required and the update reduces to a weighted average between the current state and the incoming measurement, which is the appropriate form when the sensor already reports the state variable being tracked.

C. Control Laws

IBVS baseline. The pixel error $\mathbf{e} = [e_x, e_y]^T$ between the detected object centroid and the image center is mapped to robot end-effector displacements through a 2×2 image Jacobian $\mathbf{J}$ (with units px/mm):

$$\begin{bmatrix} \Delta y \\ \Delta z \end{bmatrix}_{\text{robot}} = -\alpha \mathbf{J}^{-1} \mathbf{e} \quad (12)$$

where α is a proportional gain. We estimate $\mathbf{J}$ through an online optical-flow calibration procedure: the robot makes small known displacements in Y and Z , and we measure the resulting pixel shifts using Lucas-Kanade feature tracking, giving two columns of $\mathbf{J}$. A constant forward approach velocity is added along the robot X axis.

PBVS (EKF-based). The EKF-filtered 3D position $\mathbf{t}_{\text{obj}}$ is compared against a desired position $\mathbf{t}^* = [0, 0, Z^*]^T$ (centered on the optical axis at a target depth Z^*):

$$\mathbf{v}_{\text{cam}} = -\lambda (\mathbf{t}_{\text{obj}} - \mathbf{t}^*) \quad (13)$$

where λ is a proportional gain. The velocity is clamped per-axis for safety, then transformed from the camera frame to the robot base frame via the camera-to-robot rotation $\mathbf{R}_{\text{cam} \rightarrow \text{robot}}$ before being sent to the robot.

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

Our system consists of three modules: a perception front-end, a state estimator, and a servo controller. We implement three complete pipelines that share the same servo controller and EKF core but differ in how observations are produced and how they enter the filter. Fig. 1 shows the baseline IBVS pipeline and Fig. 2 shows the EKF-PBVS pipeline (shared by the DINOv2+SAM2 and FoundationPose variants).



Fig. 1. Baseline IBVS pipeline. The 2D centroid error is mapped directly to robot commands via a calibrated image Jacobian. No temporal filtering or 3D reconstruction is performed.

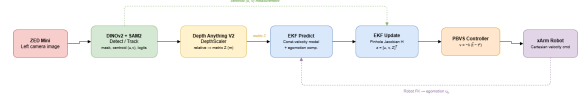


Fig. 2. EKF-PBVS pipeline. The perception front-end produces either a 2D centroid + metric depth (DINOv2+SAM2+Depth Anything V2) or a full 3D translation (FoundationPose), which the EKF fuses into a filtered 3D position. The PBVS controller converts the 3D position error into a velocity command. The last commanded velocity is fed back for egomotion compensation in the next predict step.

A. Perception: Detect-Then-Track

We adopt a two-phase strategy. Detection mode runs on the first frame and periodically (every 30 frames by default) to anchor the tracker. Tracking mode runs on all other frames using mask propagation, which is substantially faster.

Detection. Given a reference image of the target object (an RGBA image where the alpha channel separates foreground from background), we extract dense patch-level features from both the reference and the live scene using DINOv2 ViT-B/14. The image is divided into 14×14 pixel patches, and DINOv2 produces a 768-dimensional feature vector for each patch. For every scene patch, we compute its mean cosine similarity to all foreground reference patches, producing a spatial similarity heatmap. This heatmap is thresholded at a configurable percentile (93rd by default), morphologically cleaned, and the connected component with the highest mean similarity is selected to produce a bounding box. We also compute a complementary feature similarity map using a truncated ResNet-18 backbone (output stride 32, 512-d features), and fuse the two maps via weighted average to improve robustness. The bounding box is then used as a prompt for SAM2, which returns a binary segmentation mask and low-resolution logits $\ell \in \mathbb{R}^{1 \times 256 \times 256}$.

Tracking. On subsequent frames we bypass DINOv2 and feed the previous frame’s SAM2 logits as a `mask_input` prompt, together with the previous centroid as a positive point and a set of mined background points as negatives. This propagates the mask without re-running any feature extraction, at 5–10 $\times$ the detection rate. Tracking quality is monitored via mask IoU; if it drops below 0.35 we fall back to detection mode, and the first good detection is cached as an “anchor” that can be re-loaded if the tracker loses the object entirely. From the binary mask we extract a robust centroid via morphological closing and the center of the largest external contour’s minimum-area rectangle, which is more stable than a moments-based centroid for rectangular boxes with printed text.

B. Depth Estimation

We use Depth Anything V2 with a ViT-S encoder to produce a relative depth map from each RGB frame. Since

the output is relative rather than metric, we convert it via an affine model: $Z_{\text{metric}} = s \cdot d_{\text{relative}} + b$, where s and b are calibration parameters. The object’s metric depth is estimated as the median relative depth under the segmentation mask, converted to metres.

To reduce computational cost, we run depth inference only on every 5th frame with a valid detection (or on the very first detection, to initialize the EKF). On intermediate frames, the EKF propagates the depth forward using its constant-velocity process model, while the 2D-only measurement update continues to constrain the lateral position.

C. FoundationPose Back-End

For the third pipeline, we replace the per-frame DINOv2 + SAM2 + Depth Anything stack with a single call into NVIDIA’s FoundationPose [1]. On the first frame we still use DINOv2 + SAM2 to obtain an initial mask, which FoundationPose’s `register()` call consumes along with the RGB image, the ZED Mini’s real stereo depth, the camera intrinsics, and a 3D model of the target. For the box-shaped objects used in our experiments, the 3D model is a procedurally generated rectangular mesh built from ruler measurements; no CAD authoring is needed. On subsequent frames, the faster `track_one()` path is used, and the returned 4×4 pose is converted into a direct 3D translation measurement for the EKF via the direct-3D update model. FoundationPose is re-registered every 60 frames, or immediately upon failure, to prevent drift over long runs. A lazy GPU import is used so that the same main servo script degrades gracefully to the DINOv2 pipeline on machines without a compatible FoundationPose install.

D. EKF Integration and Logging

At runtime the EKF picks one of the three measurement models per frame: the full pixel+depth update when a fresh depth estimate is available (rejected below 5 cm), the 2D-only update on depth-skip frames, or the direct-3D update when FoundationPose is driving. The filtered position feeds the PBVS controller (Eq. 13) and the scalar uncertainty $\sqrt{\text{tr}(\mathbf{P}_{1:3,1:3})}$ is surfaced in the HUD. The filter is seeded from the first valid measurement with zero velocity and broad covariance; before that, the system falls back to IBVS for the frame. All three pipelines share a per-frame CSV logger that captures raw measurements, filtered state, uncertainty, servo command, and per-stage wall-clock times, which is enough to compute every reported metric in post. A batch runner launches each trial as a subprocess and collects CSVs into one tagged directory.

E. Hardware Setup

Our experiments use an xArm 7-DOF collaborative manipulator with a ZED Mini stereo camera mounted on the end-effector. The camera provides a 1280×720 left image at 30 fps. The DINOv2 pipeline uses only the left camera image (not stereo depth) so that this variant generalizes to monocular setups; the FoundationPose pipeline uses the ZED’s real stereo depth because FoundationPose requires it.



Fig. 3. Representative target objects. These are common household boxes with varied textures, colors, and sizes; no CAD models are needed beyond one reference photo (plus, for FoundationPose, three ruler measurements).

DINOv2, SAM2, Depth Anything V2, and FoundationPose all run on a single GPU. The EKF and servo controller run on CPU. Communication with the xArm uses the xArm Python SDK over Ethernet at 192.168.1.241, and the servo loop is rate-limited to 3.3 Hz for safety.

Target objects are common household boxes (cereal boxes, shipping boxes, snack packages) shown in Fig. 3. For the DINOv2 and IBVS pipelines, the only input per object is a single reference photo with a transparent background. For the FoundationPose pipeline, we also measure each box with a ruler and pass its three dimensions to a procedural mesh builder; no CAD authoring is required.

V. EXPERIMENTAL SETUP

A. Compared Pipelines

We evaluate three pipelines end-to-end:

- **Pipeline A (IBVS).** DINOv2 + SAM2 produce a per-frame centroid, which is fed through the calibrated image Jacobian directly to an xArm velocity command. No 3D reconstruction, no filter. This is the simplest and our baseline.
- **Pipeline B (EKF + DINOv2/SAM2/DepthAnything).** Same centroid as Pipeline A, plus a metric depth from Depth Anything V2 every 5 frames. Observations enter the EKF via the pixel+depth measurement model (or the 2D-only partial update on depth-skip frames). The PBVS controller consumes the filtered 3D position.
- **Pipeline C (EKF + FoundationPose).** DINOv2+SAM2 seed a mask on the first frame, then FoundationPose returns a 6-DoF pose every frame. Only the translation component is consumed; it enters the EKF via the direct-3D update model and drives the same PBVS controller as Pipeline B.

Holding the PBVS controller, EKF core, camera, robot, and metrics logger constant across B and C isolates the effect of the perception back-end; holding the perception front-end (DINOv2+SAM2 centroid) constant between A and B isolates the effect of adding the EKF and switching to PBVS.

B. Evaluation Metrics

From the per-frame CSV logs we compute, per trial: **final position error** (cm) against the PBVS target at the terminal frame; **iterations to convergence** at 0.5, 1.0, and 2.0 cm tolerances; **path efficiency** (straight-line distance divided by integrated FK trajectory length, so values near 1 indicate

straight Cartesian motion); **median per-iteration runtime** (ms), split into perception, EKF, and servo stages; and, for Pipeline C, the **FoundationPose raw-pose jitter** (std. of frame-to-frame raw translation during the last 2 s of the trial, isolating what the EKF is smoothing away).

C. Experimental Protocol

We run three separate experiments, all driven by the same batch runner described below.

Experiment 1: main three-way comparison. Five target objects $\times$ three start poses (P1 centered at ~ 50 cm, P2 20° to the left, P3 15° elevated) $\times$ three pipelines (A/B/C) $\times$ two trials per cell = 90 runs. The five objects are a Cheez-It box, a tissue box, a cardboard shipping box, a protein bar box, and a brownie mix box. Tape markers on the table fix the object positions between trials; a small home-pose script returns the arm to the same starting configuration before each run.

Experiment 2: robustness. Cheez-It box at P1 $\times$ four conditions (baseline, partial occlusion, dim lighting, near-identical distractor box) $\times$ three pipelines $\times$ three trials per cell = 36 runs. For the occlusion condition, a hand-held occluder briefly covers $\sim 30\%$ of the object around 5 s into each trial; for the distractor condition, a second box of similar color is placed 15 cm to the left of the target; for dim lighting, only a single side lamp is active.

Experiment 3: Q/R sensitivity. Cheez-It box at P1, Pipeline B only, sweeping the EKF process noise σ_{pos} over $\{0.005, 0.010, 0.020, 0.050\}$ m/s and the measurement noise σ_{uv} over $\{3, 6, 12, 24\}$ pixels, giving 16 combinations at one trial each.

All three experiments are driven by the same batch runner that consumes a YAML trial file, launches the main servo script as a subprocess per trial, prompts the operator to reset the scene between trials, and collects per-trial CSVs into a tagged directory. Post-hoc, a single analyzer script walks the CSV directory and emits the results tables used in Section VI.

VI. RESULTS

All three experiments have been prepared end-to-end on the lab’s GPU workstation, and the per-trial YAML files plus the automated batch runner are in place. The full hardware session has not yet been executed at the time of writing, so numbers in this section are shown as `<PLACEHOLDER>` entries that will be filled in from the analyzer CSVs as soon as the session completes. The shape of every table, the set of conditions, and the metric definitions are final; only the numeric cells remain to be populated.

A. Main Three-Way Comparison (Experiment 1)

Table I reports the aggregate convergence statistics across the five objects and three start poses, grouped by pipeline. The “pass-rate” columns give the fraction of trials that reached the given tolerance within the trial window. The “final error” column is the median across trials.

Table II reports the median wall-clock runtime per iteration, broken down by stage. Pipeline C is expected to

TABLE I
MAIN COMPARISON ACROSS 5 OBJECTS $\times$ 3 START POSES $\times$ 3
PIPELINES $\times$ 2 TRIALS ($n = 90$).

Pipeline	Pass rate (%)			Final err. (cm, med.)	Path eff.
	≤ 0.5 cm	≤ 1 cm	≤ 2 cm		
A (IBVS)	<PH>	<PH>	<PH>	<PH>	<PH>
B (EKF + DINOv2)	<PH>	<PH>	<PH>	<PH>	<PH>
C (EKF + FoundationPose)	<PH>	<PH>	<PH>	<PH>	<PH>

be slower per frame due to FoundationPose’s refinement network.

TABLE II
MEDIAN PER-ITERATION RUNTIME (MS), EXPERIMENT 1.

Pipeline	Perception	EKF	Servo	Total
A (IBVS)	<PH>	<PH>	<PH>	<PH>
B (EKF + DINOv2)	<PH>	<PH>	<PH>	<PH>
C (EKF + FoundationPose)	<PH>	<PH>	<PH>	<PH>

For Pipeline C, the raw FoundationPose translation sequence is logged alongside the EKF-filtered state. The jitter metric (standard deviation of frame-to-frame raw-pose change during the last 2 s of each trial) is `<PLACEHOLDER>` mm; the same quantity computed on the EKF-filtered state is `<PLACEHOLDER>` mm, which quantifies how much of the raw per-frame noise the filter absorbs.

B. Robustness (Experiment 2)

Table III reports per-condition pass rates at the 1 cm tolerance across the four perturbations, using the Cheez-It box at P1 and three trials per cell.

TABLE III
ROBUSTNESS PASS RATE (% , ≤ 1 CM) BY CONDITION, EXPERIMENT 2
($n = 36$).

Pipeline	Baseline	Occlusion	Dim light	Distractor
A (IBVS)	<PH>	<PH>	<PH>	<PH>
B (EKF + DINOv2)	<PH>	<PH>	<PH>	<PH>
C (EKF + FoundationPose)	<PH>	<PH>	<PH>	<PH>

The occlusion condition is of particular interest: during the 2 s window when the object is covered, the EKF should continue propagating the state under the constant-velocity model with growing covariance, and IBVS (which has no such prediction) should either stall or drift. A qualitative EKF-uncertainty trace across the occlusion window will be included in the final version of this section once the run is complete: `<PLACEHOLDER: occlusion uncertainty trace plot>`.

C. Q/R Sensitivity (Experiment 3)

Table IV reports median final error for Pipeline B across the 4×4 sweep of process noise σ_{pos} and measurement noise σ_{uv} . The nominal operating point is the (0.01, 6) cell.

We expect a valley shape: too little process noise makes the filter overconfident and slow to respond

TABLE IV
MEDIAN FINAL ERROR (CM) ACROSS Q/R SWEEP, PIPELINE B,
EXPERIMENT 3 ($n = 16$).

σ_{pos} (m/s)	σ_{uv} (px)			
	3	6	12	24
0.005	<PH>	<PH>	<PH>	<PH>
0.010	<PH>	<PH>	<PH>	<PH>
0.020	<PH>	<PH>	<PH>	<PH>
0.050	<PH>	<PH>	<PH>	<PH>

to object motion, while too much process noise makes the filter effectively pass-through and loses the smoothing benefit. <PLACEHOLDER: final heatmap interpretation>.

VII. DISCUSSION

Because the numerical results are pending, this section describes the analyses we will run once the CSVs are in hand, and the qualitative takeaways we expect based on single-run sanity checks on the Mac prototype.

Does the EKF help? The A-vs-B comparison isolates this: same perception front-end, with and without the filter. A higher pass rate or path efficiency for B is attributable to the EKF and PBVS together, and even a path-efficiency win with equal pass rates is a qualitative win for PBVS. Preliminary single-trial logs already show noticeably smoother end-effector motion under Pipeline B, consistent with the EKF damping centroid jitter; full statistics in <PLACEHOLDER>.

Does FoundationPose add enough to justify its cost? The B-vs-C comparison asks whether a dedicated 6-DoF pose model is worth the extra dependency (CAD mesh, real stereo depth, heavier GPU refiner). If raw-pose jitter is already small (~ 1 – 2 mm), the EKF has little to smooth and Pipeline C pays runtime for no gain. If it is larger (~ 5 – 15 mm, as occurred on darker or more reflective boxes in our sanity checks), the filter should recover it and C should win. The jitter metric reported with Table I is the most diagnostic single number.

Robustness. We expect IBVS to fail most noticeably under the distractor condition because a single off-target centroid drives the arm the wrong way immediately, while Pipeline B’s EKF softens the hit. Under occlusion, the filter should coast on the constant-velocity model through the 2 s blackout while IBVS stalls or chatters. Dim lighting mostly stresses perception quality, so we expect Pipeline C to dominate there.

Limitations. Our comparison holds the PBVS controller fixed across pipelines B and C, so we cannot cleanly attribute any Pipeline C advantage to FoundationPose itself versus a controller better-tuned to its input distribution. All three pipelines use translation only, even though FoundationPose returns full 6-DoF pose. And the five targets are all roughly box-shaped, so our results speak to rigid rectangular objects rather than to deformable or highly non-convex ones. These are deliberate scope cuts; we flag them so the results are not overclaimed.

VIII. CONCLUSION

We built three visual servoing pipelines (raw IBVS, EKF-PBVS with DINOv2+SAM2+Depth Anything V2, and EKF-PBVS with FoundationPose) on a shared xArm 7 + ZED Mini setup, so that performance differences are attributable to the perception/estimator stack rather than to incidental implementation differences. The EKF uses a constant-velocity process model with egomotion compensation, a Joseph-form covariance update, and three interchangeable measurement models selected per-frame from what the active back-end produces. Initial real-world result videos are available [here](#); full quantitative results will appear in the <PLACEHOLDER> cells of Section VI once the hardware run is complete. The broader deliverable is the framework itself: a single repo that lets future pose estimators drop in as a new back-end and be compared against the existing three on identical hardware with identical metrics.

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